



This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

Simone Scannapieco

Corso avanzato per Venis S.p.A, Venezia, Italia

Novembre 2025

Note

[illegible]



SPEECH-TO-TEXT

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, providing a template for writing or drawing. There are no margins, text, or other markings on the paper.

-  Simone Scannapieco

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

```

...
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
<dependency>
  <groupId>org.springframework.ai</groupId>
  <artifactId>spring-ai-starter-model-openai</artifactId>
</dependency>
...

```

```
spring:
  application:
    name: demo
  ai:
    openai:
      api-key: ${OPENAI_API_KEY}
```

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;

    AudioController(OpenAiAudioTranscriptionModel transcriptionModel) {
        this.openAiAudioTranscriptionModel = transcriptionModel;
    }

    @GetMapping("/transcribe")
    String transcribe(@Value("classpath:filename.mp3") Resource audioFile) {
        return openAiAudioTranscriptionModel.call(audioFile);
    }
}
```

- Simone Scannapieco
 Spring AI - Corso avanzato
 Venis S.p.A, Venezia, IT
 5 / 16

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Controller di trascrizione

```

@RestController
@RequestMapping("/api")
public class AudioController {

    ...

    @GetMapping("/transcribe-options")
    String transcribeWithOptions(@Value("classpath:filename.mp3") Resource audioFile) {
        var audioTranscriptionResponse = openAiAudioTranscriptionModel.call(new AudioTranscriptionPrompt(
            audioFile, OpenAiAudioTranscriptionOptions.builder()
                .prompt("Topics")
                .language("it")
                .temperature(0.5f)
                .responseFormat(OpenAiAudioApi.TranscriptResponseFormat.VTT).build()));
        return audioTranscriptionResponse.getResult().getOutput();
    }
}

```

- ➔ `AudioTranscriptionPrompt` incapsula la risorsa audio e le opzioni di trascrizione (`OpenAiAudioTranscriptionOptions`)
 - ➔ `.prompt` fornisce al LLM contesto riguardo gli argomenti trattati nel *file* audio
 - ➔ `.language` evita al LLM l'attività di *language guessing*
 - ➔ `.responseFormat` per calibrare il formato di *output* al contesto applicativo di utilizzo
- ⚠ con `call(audioFile)`, Spring AI si occupa di creare un *bean* `OpenAiAudioTranscriptionOptions` con i valori di *default*

Note

[illegible]

```
spring:
  application:
    name: demo
  ai:
    openai:
      api-key: ${OPENAI_API_KEY}
      audio:
        transcription:
          model: whisper-1          # whisper-1, gpt-4o-transcribe, gpt-4o-mini-transcribe
          response-format: json    # json, text, srt, verbose_json, vtt
          prompt: Topics           # Should match the audio language
          language: it             # ISO-639-1 format to improve accuracy and latency
          temperature: 0           # [0, 1]. 0 means automatically adjusting temperature
          ...
```

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- Simone Scannapieco

Note

[illegible]

```

AUDIO_PATH="path/to/sample.mp3"
MIME_TYPE=$(file -b --mime-type "${AUDIO_PATH}")
NUM_BYTES=$(wc -c < "${AUDIO_PATH}")
DISPLAY_NAME=AUDIO

tmp_header_file=upload-header.tmp

# Initial resumable request defining metadata.
# The upload url is in the response headers dump them to a file.
curl "https://generativelanguage.googleapis.com/upload/v1beta/files" \
  -H "x-goog-api-key: $GEMINI_API_KEY" \
  -D upload-header.tmp \
  -H "X-Goog-Upload-Protocol: resumable" \
  -H "X-Goog-Upload-Command: start" \
  -H "X-Goog-Upload-Header-Content-Length: ${NUM_BYTES}" \
  -H "X-Goog-Upload-Header-Content-Type: ${MIME_TYPE}" \
  -H "Content-Type: application/json" \
  -d '{"file": {"display_name": "${DISPLAY_NAME}"}}' 2> /dev/null

upload_url=$(grep -i "x-goog-upload-url: " "${tmp_header_file}" | cut -d" " -f2 | tr -d "\r")
rm "${tmp_header_file}"

# Upload the actual bytes.
curl "${upload_url}" \
  -H "Content-Length: ${NUM_BYTES}" \
  -H "X-Goog-Upload-Offset: 0" \
  -H "X-Goog-Upload-Command: upload, finalize" \
  --data-binary "@${AUDIO_PATH}" 2> /dev/null > file_info.json

file_uri=$(jq ".file.uri" file_info.json)
echo file_uri=$file_uri

```

Note

[illegible]

- 1 Utilizzo Spring Web puro (RestController, ResponseEntity, HttpEntity, ...)
- 2 Utilizzo Spring AI (STT attraverso ChatClient)

Note

- 1 Creazione interfaccia ed implementazione del servizio
- 2 Modifica controllore MVC
- 3 Test delle funzionalità con Postman/Insomnia

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, providing a guide for writing. There are no margins, text, or other markings on the paper.

```
package it.venis.ai.spring.demo.services;

import org.springframework.core.io.Resource;

import it.venis.ai.spring.demo.model.Answer;

public interface MultiModalityService {

    Answer getTranscriptionFromAudioFile(Resource audioFile);

}
```

Note

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface. There is no handwriting or other markings on the paper.

```
package it.venis.ai.spring.demo.services;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;

    }

    @Override
    public Answer getTranscriptionFromAudioFile(Resource audioFile) {

        String contentType = getContentType(audioFile);

        Media audioMedia = new Media(
            MimeTypeUtils.parseMimeType(contentType), audioFile);

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Trascrivi il seguente file audio: ").media(audioMedia))
            .call()
            .content());

    }

    ...
}
```

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

```
...
private String getContentType(Resource multimediaFile) {
    String contentType = URLConnection.guessContentTypeFromName(multimediaFile.getFilename());

    if (contentType == null || contentType.equals("application/octet-stream")) {
        String fileName = multimediaFile.getFilename();
        if (fileName != null) {
            String extension = fileName.substring(fileName.lastIndexOf('.') + 1).toLowerCase();
            contentType = switch (extension) {
                case "mp3" -> "audio/mpeg";
                case "wav" -> "audio/wav";
                case "m4a" -> "audio/mp4";
                case "ogg" -> "audio/ogg";
                case "flac" -> "audio/flac";
                default -> "media/unknown";
            };
        } else {
            contentType = "media/unknown";
        }
    }

    return contentType;
}
}
```

Note

[illegible]

```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@RequestMapping("/geminimulti-modality")
public class GeminiMultimodalityController {

    private final MultiModalityService multiModalityService;

    public GeminiMultimodalityController(MultiModalityService multiModalityService) {

        this.multiModalityService = multiModalityService;
    }

    /**
     * Endpoint per la trascrizione di file audio in testo (Speech-to-Text).
     * Utilizza un file audio predefinito dal classpath e ne estrae la trascrizione testuale
     * tramite i servizi di riconoscimento vocale di Gemini.
     *
     * @param audioFile risorsa audio dal classpath (file WAV predefinito)
     * @return Answer oggetto contenente la trascrizione del file audio
     */
    @PostMapping("/stt")
    public Answer getTranscriptionFromAudioFile(@Value("classpath:Venis_descrizione_azienza.wav") Resource audioFile) {

        return this.multiModalityService.getTranscriptionFromAudioFile(audioFile);
    }

}
```

Note

[illegible]

16 / 16

[illegible]